

Introduction to Neuropsychological (cognitive) Intervention

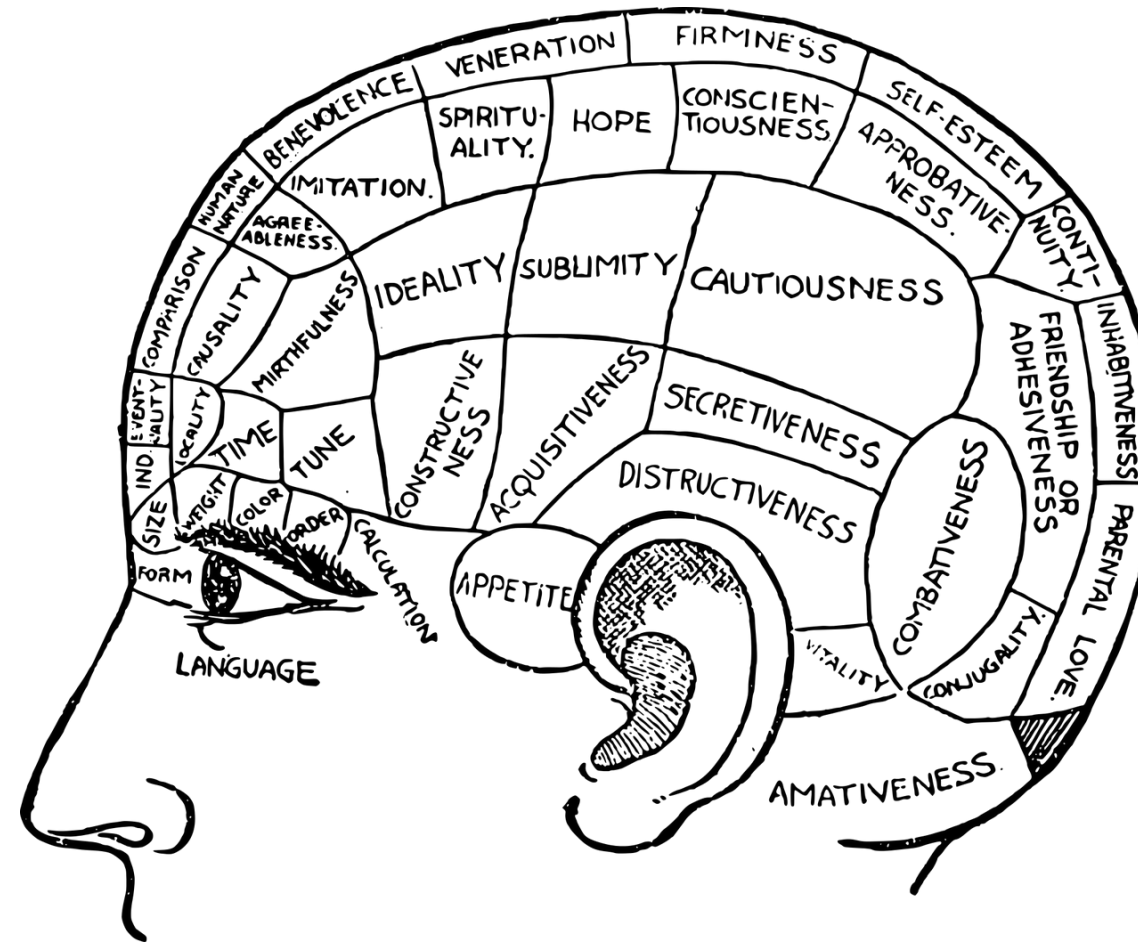
Matthew McPhee, Ph.D., C.Psych.

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Learning objectives:

1. Neuropsychology in context: understand and appreciate the historical basis of the field of neuropsychology, as it relates to cognitive rehabilitation and training
2. Cognitive training & rehabilitation:
 - a) Identify the basic tenets of each approach to intervention
 - b) Describe important limitations and open questions
 - c) Appreciate the commonalities and differences in treatment approach

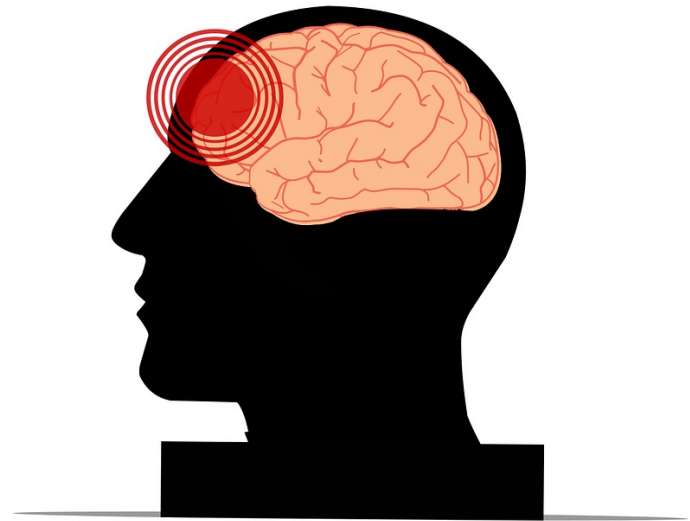
Neuropsychology in context



Neuropsychology in context

First world war: thousands of soldiers returned home with brain injuries; this laid the foundation for the specialty to grow rapidly (e.g., localization of injury)

Second world war: with another surge in the number of brain injured soldiers, neuropsychologists continued in an assessment-dominant context



Neuropsychology in context



Educational testing: standardized testing was not new; there was already wide-spread use of testing in education (e.g., Binet's intelligence tests)

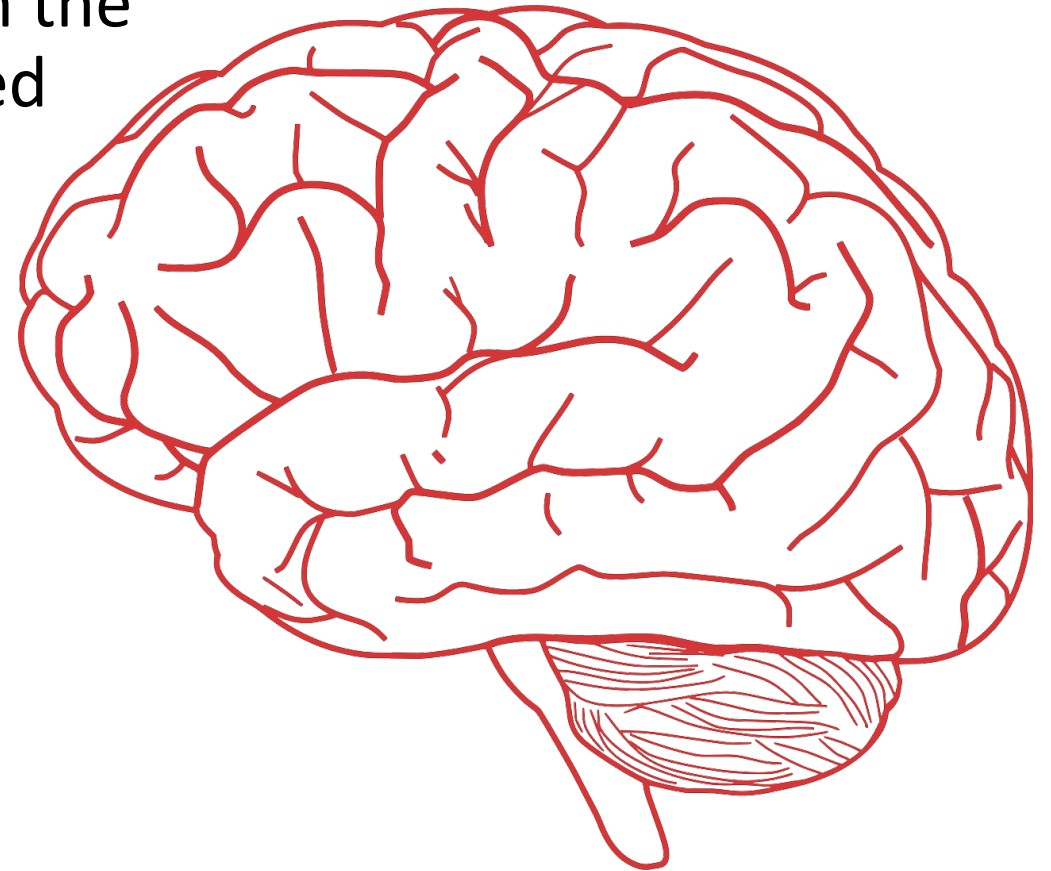
Basic research: experimental studies of brain-behavior relationship solidified the theoretical underpinnings of neuropsychological assessment

Neuropsychology in context

Neuroimaging: impressive advances in the ability to image a brain meant less need for tests to identify injured areas

New areas of focus emerged:

- Diagnosis
- Characterizing cognition
- Impact on daily life
- Treatment planning
- Treatment delivery

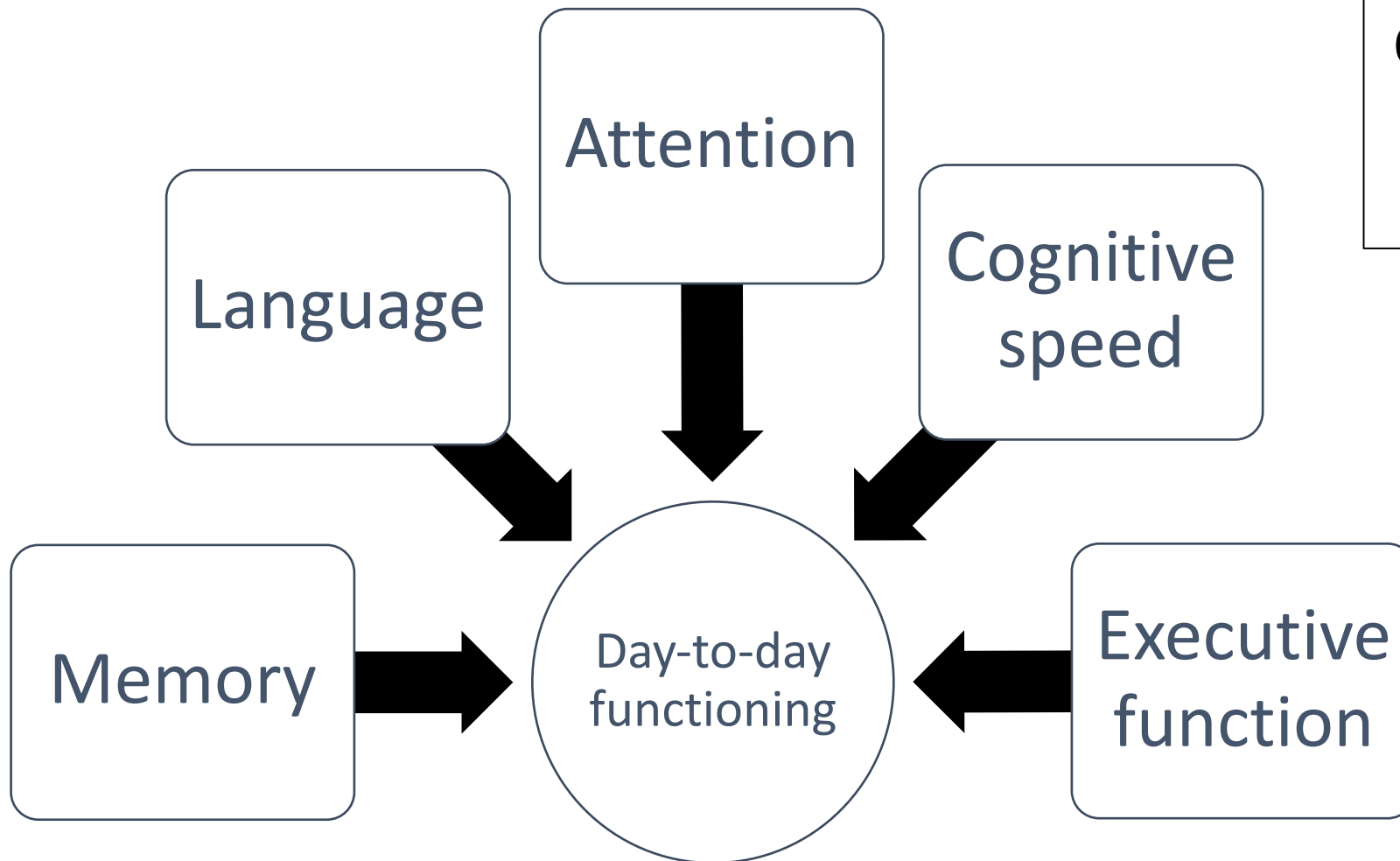


Neuropsychology in context

Historically: A client has suffered a traumatic brain injury – what impact has this had on thinking skills and where is the injury?

Contemporary: If there is an impact on thinking skills, what is the functional consequence and what can be done?

Neuropsychology in context



**Our thinking skills impact
our ability to engage in
everyday activities**

How do we help when cognitive difficulties impact daily life?

Cognitive Training

Basic approach: ‘Trainee’ completes structured cognitive tasks, typically adaptive to performance, over a period of time or sessions

- Typically grounded in a cognitive theory
- Adaptive design is key to neuroplasticity (Cicerone et al 2019)

Neuroplasticity: “the ability of the nervous system to respond to intrinsic or extrinsic stimuli by reorganizing its structure, function and connections”

(Cramer et al. 2011, p. 1591)

- A commonly-invoked mechanism for the efficacy of cognitive training

Training is typically specific to a cognitive ability, which may be a barrier to generalized rehabilitation programs (Green and Bavelier 2008)

Cicerone, K. D., et al. (2019). Evidence-based cognitive rehabilitation: systematic review of the literature from 2009 through 2014. *Archives of physical medicine and rehabilitation*, 100(8), 1515-1533.

Cramer, S. C., et al. (2011). Harnessing neuroplasticity for clinical applications. *Brain*, 134(6), 1591-1609.

Green, C. S., & Bavelier, D. (2008). Exercising your brain: a review of human brain plasticity and training-induced learning. *Psychology and aging*, 23(4), 692.

Cognitive Training

At its heart, cognitive training can be thought of as remediation, or, a restorative approach (Bahar-Fuchs et al 2013)

An area of contention/limitation: near versus far transfer

- Near: training on one task generalizes to performance on another, highly similar task (e.g., training on one working memory task improves your performance on another working memory task)
- Far: training on one task generalizes to performance on another, dissimilar or loosely similar task (e.g., working memory task training improves academic functioning or other real-world skills)

Cognitive Training – An example

Working memory training as an adjunct intervention for substance use disorder in an inpatient setting (Hendershot et al., 2018)

Cognitive training: 5 weeks (25 sessions) of adaptive auditory/verbal and visuospatial computer-based working memory training

- Adaptive feature: training either increased (longer span) or decreased (shorter span) in difficulty, based on performance

Outcomes:

- Near transfer effect: better performance on a classic working memory test
- Far transfer effect: no effect of intervention on relapse

What about a more general approach, geared to real-world functioning?

Cognitive Rehabilitation

Main emphasis: enhancing, improving, or supporting the patient's ability to meaningfully function in their day-to-day life

Main treatment target: patient-centered outcomes (i.e., goal-setting) with an emphasis on participation:

“[Cognitive rehabilitation is an] Individualized approach to helping people with cognitive impairments, in which those affected, and their families, work together with health-care professionals to identify personally relevant goals and devise strategies for addressing these” (Bahar Fuchs et al., 2013, p. 3)

Cognitive Rehabilitation

There are several approaches to cognitive rehabilitation, including:

- Compensation for cognitive deficit (emphasize and rely on strengths in place of deficits)
- Restoration of cognitive deficit
- Modification of environment
- Implementation of supports
- Systems-type interventions (e.g., family support)

Cognitive Rehabilitation

Examples of compensatory approaches:

- Internal: raising awareness of potential cognitive errors
 - For example, meta-cognitive monitoring in Goal Management Training®
- External: lists and record keeping to support memory
 - For example, memory organizer in Learning the Ropes for Living with MCI®

What about transfer effects?

- In many cases, far-transfer *is* the focus of treatment
 - For example, improving ability to remember upcoming medical appointments via consistent use of a calendar application

Cognitive Rehabilitation – An example

Errorless learning of everyday tasks in people with dementia (de Werd et al., 2013)

Rehabilitation focus: errorless learning models (e.g., modeled approach, graded approach, vanishing cues, spaced retrieval) to learn everyday tasks (e.g., use of kitchen appliances)

Outcomes: increased independent skill application

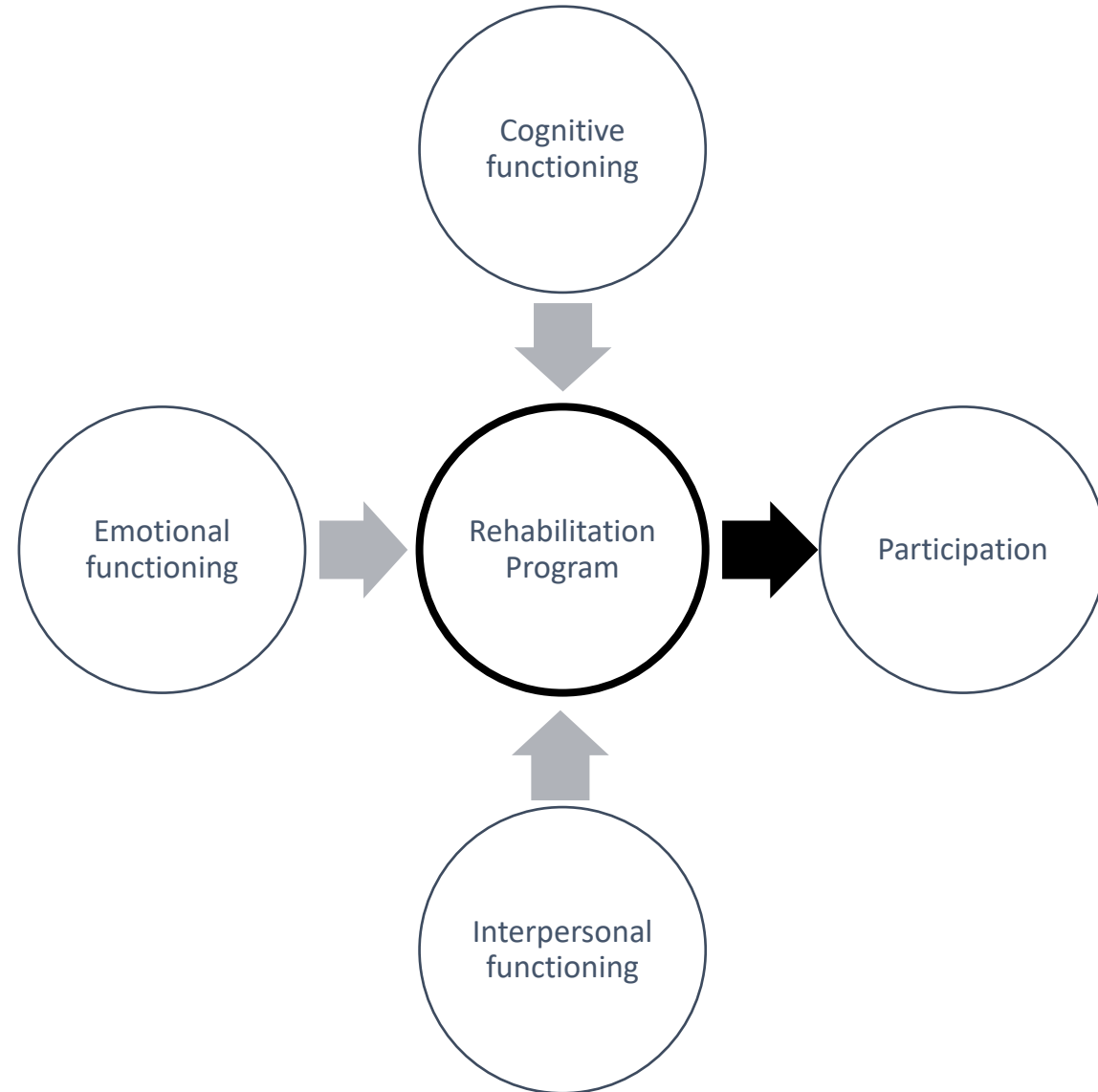
Rehabilitation versus Training

Table 1. Selected differences between cognitive training and cognitive rehabilitation

	Cognitive training	Cognitive rehabilitation
Target	Impairment	Participation restriction
Context	Structured tasks and environments	Real-world setting
Focus of intervention	Isolated cognitive abilities and processes	Groups of cognitive abilities and processes required to perform everyday tasks
Format	Individualized or group	Individualized
Proposed mechanism of action	Mainly restorative; sometimes combined with psychoeducation and strategy training	A combination of restorative and compensatory approaches combined with psychoeducation and strategy training
Goals	Improved or maintained ability in specific cognitive domains	Performance and functioning in relation to collaboratively set goals

Note: although treated as mutually exclusive, cognitive training and rehabilitation can be seen as complementary, targeting different aspects of recovery

Holistic
approaches
emphasizing
multiple domains
of functioning
(example,
biopsychosocial
model)



Complementary, not incompatible

Cognitive training and rehabilitation both offer approaches to provide treatment or intervention to individuals with varying levels of cognitive difficulties

The primary differences are in their targeted outcomes and approaches, but can often be completed in combination (these are not mutually exclusive approaches)

Newer models emphasize holistic, integrative approaches

As a newer area of clinical practice, continuous improvements, adaptations, and research will continue to shape current approaches